Design and Experimental Validation of a Portable RF Comb Generator for EMC Laboratory Applications

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***Abstract*—Portable and low-cost reference sources are valuable tools for electromagnetic compatibility (EMC) laboratories, educational activities, and experimental radio-frequency (RF) validation tasks. However, many available solutions either require complex laboratory infrastructure or lack the flexibility, portability, and autonomy needed for practical field and academic use. This work presents the design, implementation, and experimental validation of a portable RF comb generator intended for EMC laboratory applications. The proposed system integrates a programmable frequency source, a high-speed RF switching stage, embedded control, and a rechargeable battery subsystem within a compact and autonomous platform. Broadband harmonic generation is achieved through fast pulse transitions, enabling the verification of EMC measurement setups, evaluation of RF subsystems, and educational experimentation. Experimental results demonstrate sub-nanosecond switching performance, with measured fall times of approximately 244 ps, leading to harmonic content extending into the GHz range for a 10 MHz operating frequency. Time-domain and frequency-domain measurements confirm the stability and repeatability of the generated signals, while preliminary EMC characterization validates the suitability of the device for laboratory operation. The resulting platform provides an accessible, reproducible, and low-cost solution for EMC setup verification, RF experimentation, and engineering education, demonstrating the feasibility of integrating portable broadband harmonic generation capabilities into a compact standalone instrument. Index Terms— electromagnetic compatibility (EMC), RF comb generator, harmonic generation, broadband signals, EMC measurements, portable instrumentation, embedded systems, engineering education.**

***Index Terms*—electromagnetic compatibility (EMC), RF comb generator, harmonic generation, broadband signals, RF measurements, EMC laboratory, embedded control, pulse generator.**

I. Introduction

The verification and validation of measurement systems in electromagnetic compatibility (EMC) environments is essential to ensure the reliability of experimental results. In particular, radiated emissions (RE) and conducted emissions (CE) testing require the use of reference sources to periodically assess the correct performance of the entire measurement chain, including antennas, cables, preamplifiers, and spectrum analyzers [1]–[3].

Reliable reference sources are therefore fundamental not only for laboratory verification tasks but also for educational activities and experimental RF characterization.

In this context, RF comb generators have become widely used tools in EMC laboratories due to their ability to generate multiple harmonic components simultaneously in a stable and reproducible manner [4]. An RF comb generator produces a periodic signal with fast transitions in the time domain, which translates into a set of evenly spaced spectral components in the frequency domain, corresponding to multiples of the fundamental frequency. This characteristic makes them suitable for system verification, instrumentation calibration, and frequency response evaluation of RF devices [5]. Additionally, they are particularly useful for rapid validation of test setups, enabling the detection of deviations within the measurement chain [6]–[8].

Their simplicity, repeatability, and multifrequency spectral content also make them attractive for engineering education and experimental RF laboratories, where broadband signal sources are often required.

From a technological perspective, harmonic generation in RF systems relies on nonlinear elements capable of transforming a fundamental signal into a wideband frequency spectrum. Among the most commonly used techniques are step recovery diode (SRD)-based circuits [9], nonlinear transmission lines (NLTL) [10], [11], and high-speed digital logic approaches [12], each offering different trade-offs in terms of complexity, bandwidth, and cost. In particular, achieving sub-nanosecond transition times is critical for extending harmonic content into the microwave frequency range.

In recent years, the development of compact electronic platforms has enabled the emergence of low-cost portable devices for both laboratory and educational applications [5], [6]. However, many of these implementations exhibit limitations in terms of system integration, programmability, or portability, which restricts their applicability in real experimental scenarios. Consequently, there is a need for compact and reproducible solutions that combine broadband harmonic generation, embedded control, and battery-powered operation within a single platform.

This work was developed as part of an undergraduate Electronic Engineering project focused on the design, implementation, and validation of an experimental instrument for EMC testing environments. In particular, a portable RF comb generator with embedded control is proposed, capable of generating broadband harmonic spectra through a compact architecture powered by an autonomous battery subsystem. The proposed system integrates a programmable frequency source, a high-speed RF switching stage, and embedded control electronics, providing a practical platform for EMC measurement setup

verification, RF experimentation, and engineering education.

Experimental results demonstrate the ability of the system to generate sub-nanosecond transitions and broadband harmonic content, validating its applicability as a practical tool for EMC laboratory environments [7], [13].The remainder of this paper is organized as follows: Section II reviews the state of the art. Sections III and IV describe the proposed architecture and its implementation. Section V presents the experimental characterization of the prototype, while Section VI reports conducted and radiated emissions measurements performed in an EMC facility. Finally, Sections VII and VIII present the conclusions and future work, respectively.

The proposed development follows a complete engineering lifecycle, including state-of-the-art review, requirements definition, system design, implementation, verification, and validation in an operational EMC environment.

The main contributions of this work are threefold: 1. The design and implementation of a compact battery-powered RF comb generator integrating signal generation, embedded control, and autonomous operation. 2. The experimental validation of broadband harmonic generation through sub-nanosecond pulse transitions, achieving measured fall times of approximately 244 ps. 3. The demonstration of a low-cost and reproducible platform suitable for EMC laboratory verification, RF experimentation, and engineering education.

II. State of the Art

RF comb generators have been extensively studied for broadband harmonic generation and their application in RF measurement, calibration, and electromagnetic compatibility (EMC) systems. These devices generate periodic signals with fast transitions that produce evenly spaced harmonic components over a wide frequency range, enabling their use as reference sources for instrumentation verification, frequency response characterization, and EMC setup validation [4], [5].

One of the most widely adopted techniques for harmonic generation is based on step recovery diodes (SRDs), which exploit highly nonlinear charge-storage effects to generate narrow pulses and high-order harmonics efficiently [9]. SRD-based circuits are commonly employed in frequency multipliers and RF instrumentation, achieving bandwidths extending into the GHz range. However, they typically require careful impedance matching, dedicated biasing networks, and extensive circuit optimization.

Another well-established approach employs nonlinear transmission lines (NLTLs), which generate ultrafast transitions through distributed nonlinear effects [10], [11]. These structures can achieve picosecond-scale pulse widths and extremely wide harmonic spectra, making them suitable for demanding applications such as ultrawideband systems and high-resolution radar. Nevertheless, NLTL implementations generally involve greater design complexity and higher cost.

To reduce implementation complexity and cost, several authors have explored high-speed digital logic techniques based on ECL and PECL devices [12]. These architectures can produce sub-nanosecond transitions using commercially available components, providing a practical compromise between spectral performance, cost, and implementation effort.

More recently, compact and portable comb generator systems have been proposed for rapid validation of EMC setups and onsite measurements [6], [8]. These solutions facilitate verification of measurement chains and reduce testing time in laboratory and field environments.

Despite the variety of reported implementations, most existing solutions prioritize either maximum spectral performance through specialized RF structures [9]–[11] or simplified low-cost operation with limited configurability and autonomy [6], [12]. Consequently, relatively few systems combine broadband harmonic generation, embedded control, rechargeable battery operation, compact mechanical integration, and EMC-oriented functionality within a single platform.

In this context, the proposed work is positioned within the class of digitally-assisted comb generators based on high-speed switching techniques. Rather than pursuing maximum spectral performance, the proposed architecture focuses on providing a practical, portable, and reproducible instrument for EMC laboratory verification, RF experimentation, and engineering education, integrating programmable operation, embedded control, and autonomous battery-powered functionality within a low-cost platform.

TABLE I
Comparison with representative RF comb generator implementations

Work	Tech.	Port.	Batt.	Prog.	EMC
Martin (1988)	Analog	No	No	No	No
Reader (2008)	Comb Gen.	No	No	No	Partial
Kolodziej (2016)	Compact RF	Yes	No	Limited	Partial
Chia (2017)	PECL Logic	Partial	No	Limited	No
This Work	RF Switch	Yes	Yes	Yes	Yes

Table I summarizes representative RF comb generator implementations reported in the literature and compares them with the proposed system. Representative works were selected to cover analog, compact RF, and digital logic-based implementations reported in the literature. The comparison is qualitative and based on publicly available design characteristics, including portability, battery operation, programmability, and intended EMC applicability.

III. Proposed System

The proposed system consists of a portable RF comb generator designed for laboratory applications in electromagnetic compatibility (EMC). The architecture integrates signal generation, embedded control, and an autonomous power subsystem into a single compact platform, enabling portable standalone operation.

The core functionality of the system is based on the generation of a periodic signal from a programmable frequency source. This signal is subsequently processed by a high-speed switching stage, producing short-duration pulses with fast transition times. These sharp transitions result in a wideband harmonic spectrum in the frequency domain,

which is characteristic of RF comb generators. A high-speed RF switching approach was selected because it provides a favorable compromise between implementation complexity, cost, and harmonic generation capability when compared with alternative techniques such as SRD-based or NLTL-based architectures.

The system architecture was defined to satisfy four primary design objectives:

Broadband: Harmonic generation suitable for EMC verification activities.
Portable: Operation without external infrastructure.
Programmable: User-configurable frequency generation.
Low-cost: Implementation using commercially available components.

These objectives guided the selection of the signal generation architecture, power subsystem, and enclosure design.

Portability was considered a fundamental design requirement, motivating the integration of battery-powered operation and compact mechanical packaging. This enables standalone use in EMC laboratories, educational environments, and field demonstrations without requiring external power supplies or dedicated infrastructure.

From a structural perspective, the system is organized into three major subsystems:

Control subsystem: Includes the USB communication interface and the embedded microcontroller, responsible for system configuration, frequency programming, monitoring, and user interaction.

Signal generation subsystem: Comprises the programmable clock generator, RF pulse generation stage, and output matching network, responsible for generating and conditioning the wideband harmonic spectrum.

Power subsystem: Includes the rechargeable battery pack and power regulation circuitry, providing autonomous operation and the required supply voltages for all system blocks.

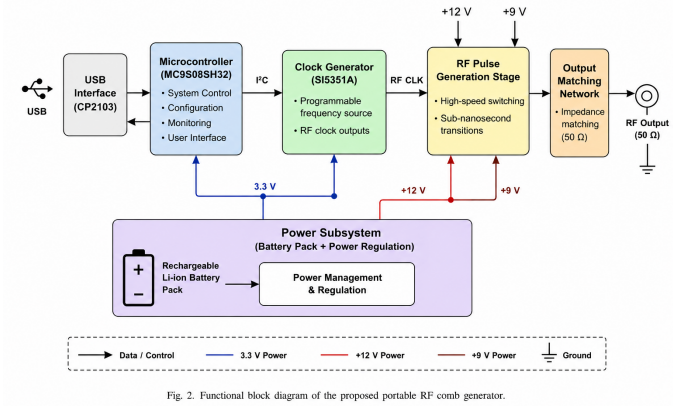


Fig. 2. Functional block diagram of the proposed portable RF comb generator.

Fig. 1. Functional block diagram of the proposed RF comb generator, showing the interaction between the control, signal-generation, and power subsystems. The programmable frequency source drives the high-speed RF switching stage responsible for broadband harmonic generation, while the embedded controller provides configuration and monitoring capabilities.

The programmable frequency generator provides a configurable fundamental signal that determines the spacing between the generated harmonics. This signal is then applied to the RF pulse generation stage, where the required nonlinear transformation takes place to produce high-speed transitions and, consequently, a wide spectral content. Fig. 1 presents the block diagram of the system, illustrating the interaction between the main functional components and the signal flow across the architecture.

The embedded control system enables configuration of operating parameters, including frequency selection and RF output activation. A serial communication interface is also implemented to facilitate user interaction and system monitoring.



Fig. 2. Mechanical implementation of the portable RF comb generator. The enclosure integrates the signal-generation electronics, embedded control subsystem, rechargeable battery pack, and interchangeable radiating antenna used during radiated emissions measurements. Comb generator with antenna (for RE) and without it (for CE)

Fig. 2 shows the mechanical implementation of the system, integrated into a compact enclosure that enables direct use in laboratory environments without requiring additional infrastructure. This feature is particularly relevant in EMC applications, where portability and ease of deployment are key factors.

The power subsystem is based on rechargeable batteries, providing full autonomy and eliminating the need for external power sources during operation. This is particularly relevant in EMC testing, as it avoids interference associated with power cables.

Finally, the system is integrated into a compact mechanical structure that facilitates transportation and deployment in

laboratory environments. This integration results in a fully autonomous test device suitable for both educational purposes and experimental validation in controlled EMC setups.

The resulting architecture integrates signal generation, embedded control, autonomous power supply, and compact mechanical implementation into a single portable platform. This enables practical use in EMC verification activities, RF experimentation, and engineering education while maintaining low implementation complexity and design reproducibility.

IV. System Implementation

This section describes the implementation of the proposed system, including the electronic design, power subsystem, and mechanical integration. The implementation aims to achieve a compact, portable, and reproducible solution suitable for EMC laboratory environments.

The printed circuit board (PCB) was designed using a double-layer FR4 substrate in order to achieve a low-cost, easily manufacturable, and reproducible implementation. Although higher-performance microwave substrates are available, FR4 was selected due to its widespread availability and adequate electrical performance within the targeted frequency range. The layout follows high-frequency design criteria, including minimized critical RF trace lengths, continuous ground planes, and via stitching techniques to improve current return paths and reduce parasitic effects.

Fig. 4 shows the PCB top layer, highlighting the component distribution and RF routing strategy used to preserve signal integrity.

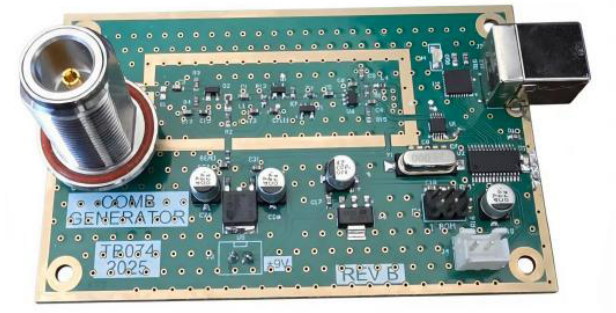


Fig. 3. Top-layer PCB layout of the RF comb generator showing component placement, RF routing strategy, grounding structure, and compact integration of the signal-generation, control, and power subsystems.

The RF pulse generation stage represents the core of the system. In this stage, the signal generated by the programmable frequency source is processed using a high-speed switching topology that enables the generation of pulses with sub-nanosecond transition times. This behavior is essential for broadband harmonic generation, as the sharpness of the signal edges directly determines the high-frequency spectral content.

The system incorporates a USB communication interface that supports both configuration and monitoring tasks. This approach enhances usability and integration in different experimental scenarios. The power subsystem is based on a rechargeable lithium-ion battery pack connected in series to provide the required operating voltage. A dedicated charging and balancing module is included to ensure safe and reliable operation.

Fig. 5 presents the internal assembly of the power subsystem, including the battery pack and regulation modules.

Fig. 6 shows the schematic of the RF pulse generation stage. This stage combines a high-speed RF buffer amplifier and discrete RF transistors to generate fast-transition pulses with rich harmonic content. The first transistor operates in switching mode to produce sharp edges, while the second stage provides gain and output drive capability. Passive matching and filtering networks were optimized to improve harmonic distribution and output impedance matching.

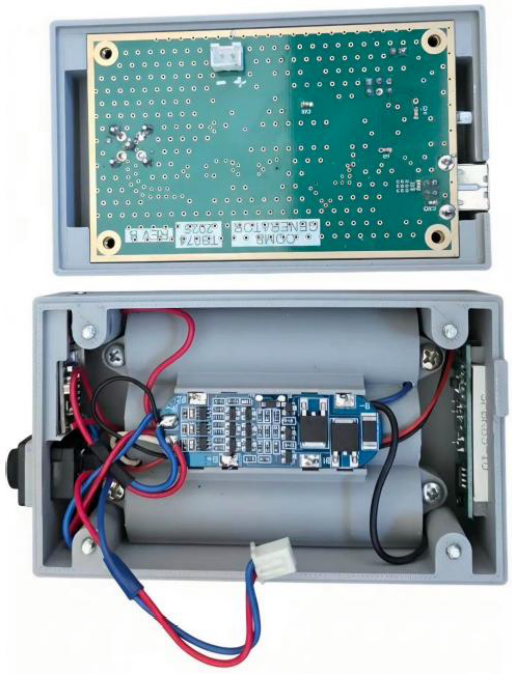


Fig. 4. Internal assembly of the autonomous power subsystem, including the rechargeable battery pack, charging circuitry, balancing electronics, and voltage regulation modules used to support standalone operation

The current consumption of the system depends on the selected operating frequency, increasing with higher switching activity in the RF stage. This behavior is consistent with high-speed pulse generation systems, where power dissipation is related to rapid charge and discharge processes.

From a mechanical perspective, the system is integrated into a portable enclosure designed to minimize internal interference and preserve signal integrity. The internal arrangement of components has been optimized to reduce coupling effects and maintain stable operation.

Overall, the implemented system provides a compact, autonomous, and configurable solution suitable for measurement and validation tasks in EMC environments.

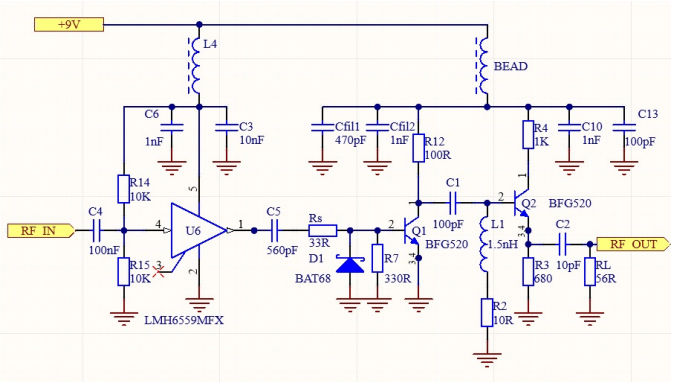


Fig. 5. RF pulse generation stage. The circuit combines a high-speed buffer amplifier, switching transistors, and output matching networks to generate fast-transition pulses and broadband harmonic content.

TABLE II
MAIN SYSTEM SPECIFICATIONS

Parameter	Value
Fundamental frequency / harmonic spacing	10–50 MHz
Power supply	Battery-operated (3S Li-ion)
Current consumption	110–160 mA
Estimated autonomy	16–20 h
Measured fall time	244 ps
RF output impedance	50 Ω
PCB technology	2-layer FR4
Communication interface	USB (CP2103)

V. EXPERIMENTAL RESULTS

The experimental validation of the proposed system was carried out through time-domain and frequency-domain measurements, aiming to characterize the generated pulses and the resulting harmonic content. These measurements allow evaluating the system's capability to generate broadband signals suitable for EMC testing applications.

A. Time-Domain Analysis

Time-domain characterization was performed using a high-speed oscilloscope to observe the output waveform of the system.

Fig. 6 shows the measured signal for a 10 MHz operating frequency, where a periodic pulse train can be observed. The waveform exhibits stable and repetitive behavior, which is characteristic of pulse-based harmonic generators.

Fig. 7 presents a zoomed view of the falling edge of the signal. A fall time of approximately 244 ps is measured, indicating sub-nanosecond transition performance. This fast transition time is consistent with the requirements for broadband signal generation, as the sharpness of the signal edges directly determines the extension of high-frequency spectral components.

Fig. 8 shows the full repetitive waveform over multiple cycles, highlighting the temporal stability and consistency of the generated pulses. The low variability and repeatability of the signal confirm the robustness of the system, making it suitable as a reference source in EMC measurements.

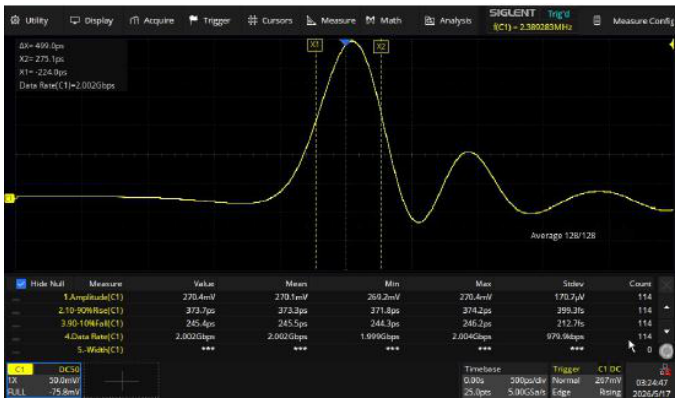


Fig. 6. Measured output waveform at a 10 MHz operating frequency, showing stable pulse generation and periodic operation of the RF comb generato



Fig. 7. Detailed view of the RF output falling edge. The measured fall time of approximately 244 ps demonstrates sub-nanosecond switching performance required for broadband harmonic generation



Fig. 8. Multiple-cycle view of the generated pulse train, illustrating waveform repeatability, timing consistency, and stable operation over successive cycles

B. Frequency-Domain Analysis

The frequency-domain behavior of the system was analyzed using a spectrum analyzer to observe the harmonic distribution.

Fig. 12 presents the measured spectrum for a 10 MHz fundamental frequency. Multiple evenly spaced harmonic components are observed, with spacing corresponding to the fundamental frequency.

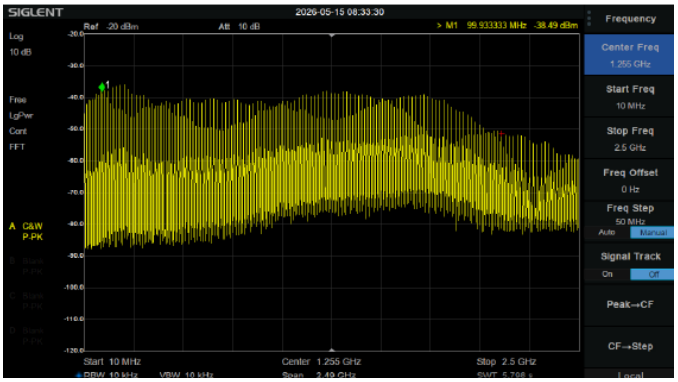


Fig. 9. Measured harmonic spectrum generated from a 10 MHz fundamental frequency. Multiple evenly spaced harmonics are observed, extending into the GHz range as a consequence of the fast pulse-transition characteristic.

The spectrum exhibits a relatively continuous envelope, with harmonic content extending into the GHz range. This result is consistent with the sub-nanosecond transition times observed in the time-domain measurements.

The obtained spectral distribution validates the operation of the device as an RF comb generator, confirming its suitability for applications requiring simultaneous availability of multiple reference frequencies.

VI. Emissions Measurements

To validate the suitability of the proposed RF comb generator as a reference source for EMC laboratory activities, both conducted emissions (CE) and radiated emissions (RE) measurements were performed.

A. Conducted emissions

Conducted emissions measurements were carried out to characterize the spectral content generated at the output of the device.

Fig. 11 Shows conducted emissions (CE) measurement setup.

The RF output of the comb generator was directly connected to the spectrum analyzer input through a 50 Ω coaxial cable, enabling direct spectral characterization of the generated harmonics while minimizing propagation losses and external coupling effects.

Conducted emissions characterization was performed by directly coupling the RF output of the comb generator to the measurement instrumentation using a 50 Ω coaxial connection. Preliminary observations confirmed the presence of harmonic components spaced according to the programmed fundamental

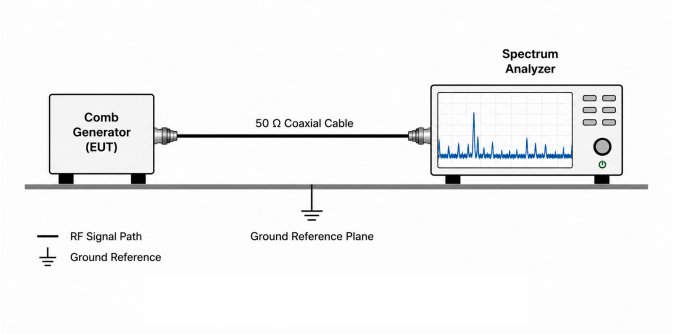


Fig. 10. Conducted emissions measurement setup. The RF output of the comb generator is directly connected to a spectrum analyzer through a 50 Ω coaxial cable, enabling characterization of the generated harmonic spectrum

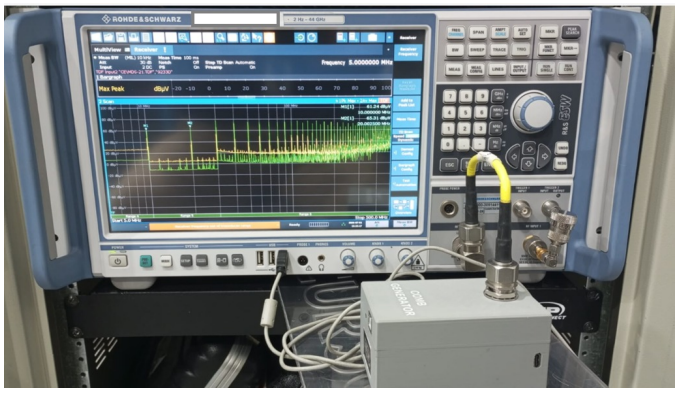


Fig. 11. Measured conducted emissions spectrum showing harmonic components spaced according to the programmed fundamental frequency, validating the broadband operation of the proposed RF comb generator.

frequency, extending into the GHz range. Fig 10 confirm the presence of harmonic components spaced according to the programmed fundamental frequency, extending into the GHz range. A complete quantitative spectral analysis of the acquired dataset is currently under post-processing and will be presented in future work.

These conducted emissions measurements validate the ability of the system to generate a stable and broadband RF comb suitable for EMC applications.

B. Radiated emissions

Radiated emissions measurements were conducted in a calibrated semi-anechoic chamber (SAC) following CISPR 32 guidelines [14], with the objective of evaluating the electromagnetic radiation produced by the device in free-space conditions.

Fig. 13 shows the radiated emissions measurement setup.

The spectrum analyzer is configured with peak and quasi-peak detectors, as required by the CISPR 32 standard.

Radiated emissions measurements confirmed the detectability of harmonic components generated by the proposed device inside the semi-anechoic chamber. Since the complete radiated emissions dataset is still under post-processing, the detailed

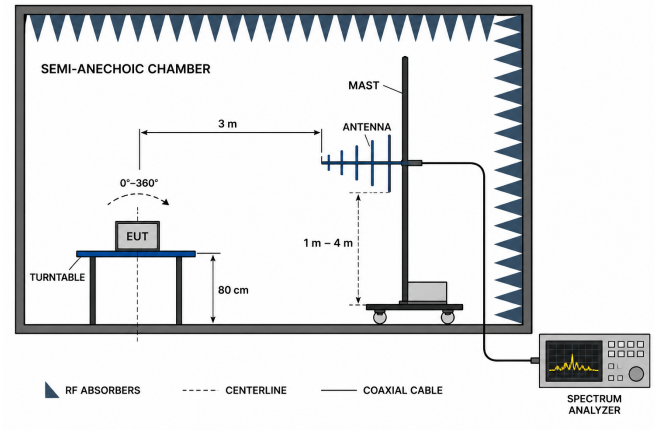


Fig. 12. Radiated emissions measurement setup used inside the semi-anechoic chamber, including the RF comb generator, receiving antenna, and spectrum analyzer configured according to CISPR 32 measurement practices

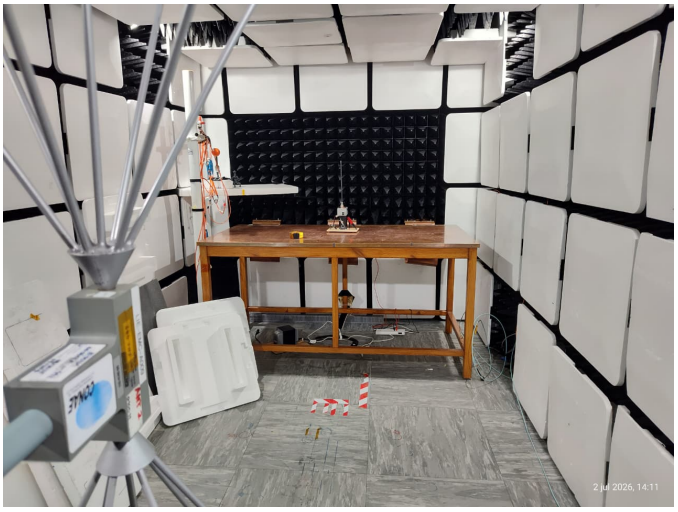


Fig. 13. Radiated emissions measurement configuration used for validation inside the semi-anechoic chamber, including the transmitting device, receiving antenna, and acquisition instrumentation.



Fig. 14. Measured radiated emissions spectrum obtained at a 3 m separation distance inside the semi-anechoic chamber. The detected harmonic components confirm the suitability of the device as a practical EMC laboratory reference sourc

quantitative analysis, including calibration and uncertainty evaluation, will be presented in future work.

These preliminary results indicate that the proposed device can be used as a laboratory signal source and as a candidate reference source for periodic verification of EMC measurement setups and anechoic chambers [8], [15].

In summary, the system achieves a suitable balance between performance, cost, and integration, making it a practical solution for both academic and laboratory EMC applications, while leaving room for further improvements in metrological characterization and frequency range extension.

VII. Conclusions

This work presented the complete development cycle of a portable RF comb generator for electromagnetic compatibility (EMC) laboratory applications, including state-of-the-art analysis, requirements definition, system design, implementation, verification, and experimental validation.

Based on the identified needs of EMC laboratories and educational environments, a compact and autonomous architecture was specified and developed, integrating a programmable frequency source, a high-speed RF switching stage, embedded control electronics, and a rechargeable battery subsystem within a single portable platform.

The implemented prototype was experimentally verified through time-domain and frequency-domain measurements. The obtained results demonstrated sub-nanosecond switching performance, achieving a measured fall time of approximately 244 ps and generating a broadband harmonic spectrum extending into the GHz range. These results confirmed compliance with the functional requirements established during the design phase.

The system was subsequently validated through conducted and radiated measurements performed in a semi-anechoic electromagnetic compatibility facility. The obtained results demonstrated the suitability of the proposed device as a practical tool for EMC measurement setup verification, RF experimentation, and educational activities.

Beyond the achieved technical performance, the project demonstrated the feasibility of developing a low-cost and reproducible RF comb generator using commercially available components while maintaining portability, autonomous operation, and ease of deployment. The developed instrument is currently being employed as a support tool in EMC laboratory activities, providing a practical source for verification and experimental characterization tasks.

Overall, the presented work demonstrates a complete engineering process, from requirements identification to operational validation, resulting in a functional and validated instrument that addresses the needs of EMC laboratories, RF experimentation environments, and engineering education.

VIII. Future Work

Although the developed system successfully fulfilled the established requirements and demonstrated its usefulness as

a practical EMC laboratory tool, several opportunities for improvement have been identified.

Future work will focus on the development of a second-generation comb generator with enhanced spectral performance and broader applicability to EMC measurement activities. Planned improvements include optimization of the RF output stage, impedance matching networks, and PCB layout to improve harmonic amplitude flatness and extend usable frequency coverage.

Another area of development will investigate the integration of a calibrated radiating structure, enabling the device to operate not only as a conducted reference source but also as a portable radiated reference source for periodic verification of measurement setups and semi-anechoic chambers.

A comprehensive metrological characterization of the instrument will also be performed. This effort will include short-term, medium-term, and long-term stability analyses, repeatability assessments, environmental sensitivity studies, and uncertainty estimation. The objective is to evaluate the suitability of the comb generator as an internal laboratory reference standard for periodic verification of EMC measurement chains, instrumentation health monitoring, and facility maintenance activities.

Finally, the experience gained during the development, validation, and operational use of the current prototype will be incorporated into an improved platform intended for wider deployment in EMC laboratories, educational environments, and experimental RF facilities.

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